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SPBEBOT, A Chatbot for Assessors Evaluating Indonesia Electronic-Based Government Systems (SPBE) Using Retrieval-Augmented Generation

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ABSTRACT

Electronic-Based Government System (SPBE) has emerged for Indonesia's digital transformation in public services, as regulated by Presidential Regulation No. 95 of 2018. The implementation of SPBE is yearly monitored and evaluated by assessors based on the Regulation of the Ministry of Administrative and Bureaucratic Reform (Kementerian PANRB) No. 3 of 2024 regarding the "Pedoman Tata Cara Pemantauan dan Evaluasi SPBE." This guideline defines an evaluation framework consisting of four domains (Policy, Governance, Management, and Services), 8 aspects, and 47 indicators, each assessed across 5 maturity levels and followed by documentary evidence. Due to the hierarchical structure and large volume of regulatory documents, assessors still rely on manual processes to retrieve information and interpret assessment criteria, resulting in time consuming evaluation activities. This research develops SPBEBOT, a chatbot with Vector Retrieval-Augmented Generation approach to support SPBE assessors in accessing and interpreting evaluation guidelines. The system implements hierarchical document splitting, parent document retrieval, multi-query expansion, and prompt routing to provide the SPBE information and audit assessment tasks. This system performance is evaluated using the Retrieval-Augmented Generation Assessment (RAGAS) which achieves 83% Context Recall, 68% Context Precision, 72% Faithfulness, and 81% Answer Relevancy. Additionally, an overall user satisfaction score of 84.6% was obtained from testing conducted by external SPBE assessors. This research can accelerated information retrieval in the SPBE evaluation process and improving assessor productivity.

Keywords- SPBE, E-Government, Auditing, Chatbot, Retrieval-Augmented Generation, Large Language Models.

I. INTRODUCTION

Improvements in the quality of public services by the government through digital transformation can be achieved with good E-Government management. The implementation of E-Government has been proven to accelerate administrative processes, reduce service costs, and increase public satisfaction [1], [2]. In Indonesia, the standards for E-Government implementation are regulated in the Electronic-Based Government System (SPBE) in accordance with Presidential Regulation No. 95 of 2018 to realize a clean, effective, transparent, and accountable government in the delivery of public services [3]. However, from the 2024 United Nations E-Government survey show that Indonesia ranks 64th between of 193 countries in the Very High E-Government Development Index (VHEGDI) category with a score of 0.7991. This indicating that the implementation of SPBE still faces various structural and operational challenges [4].

The monitoring and evaluating of SPBE implementation at the national level is the responsibility of the Ministry of Administrative and Bureaucratic Reform of the Republic of Indonesia (KemenPAN-RB), as regulated in PANRB Regulation Number 59 of 2020 concerning SPBE Monitoring and Evaluation [5]. This regulation resulted in the SPBE index being used as an indicator of the maturity of digital transformation by the government that audited by internal and external assessors. From 2023 report, the national SPBE index reached an average score 2.79 of 5 are in the "Good" category, but there were several

specific domains, such as Governance (score of 1.66) and Management (score of 2.29) that were still below the index target of 2.60 [6]. This gap in scores of the Governance and Management domains indicates that the SPBE implementation is still below standard and could affect public services to the community. This requires evaluation and audit by assessors so that recommendations for improvement can be made [7].

In the monitoring and evaluation process, assessors refer to PANRB Ministerial Regulation No. 3 of 2024 concerning "Pedoman Tata Cara Pemantauan dan Evaluasi SPBE." These audit guidelines provide a framework for assessing the SPBE index based on 4 domains (Policy, Governance, Management, Services), 8 aspects, and 47 indicators to measure the maturity level from 1 to 5 based on evidence documents. Assessors carry out a series of activities ranging from self-assessment, document verification, interviews, and field visits [6]. The complexity of the guideline documents including criteria metrics and evidence requirements for each level, makes the audit process for an assessor dependent on manual document retrieval [8]. Moreover, searching for specific evaluation information and correlating data between indicators takes a long time and prone to human error and subjective bias that will affect the consistency, accuracy, and results of the audit [9], [10].

Several studies show that the use of artificial intelligence, particularly Large Language Models (LLMs), has the potential to assist the audit process as a tool for extracting information from documents more efficiently and increasing user satisfaction by integrating it into chatbots [11], [12], [13]. Research by Emett S [12] shows that the use of LLM to assist in the internal audit process of multinational companies can streamline tasks by 50-80%. However, the use of LLM has risks and limitations when used in specific cases and requires a high level of domain knowledge [14]. There are two basic problems, namely hallucination, where the model confidently provides answers that appear factual but contain incorrect or fabricated information [15]. And secondly, the limitations of knowledge in LLM because the public data contained in the static model is not connected to the ever-evolving knowledge [16]. As a result of the above problems, using LLM alone cannot be trusted when making it an assistant in audit cases with high domain knowledge requirements and specific tasks in the context of SPBE evaluation because it can produce unacceptable inaccuracies in answers.

The Retrieval-Augmented Generation (RAG) approach is mentioned in several studies as being able to overcome the limitations of LLM knowledge, connecting it through a mechanism of retrieving document information embedded in a vector database, so that the answers provided by the chatbot are always based on evidence from valid guidelines and can be dynamically updated [17], [18], [19]. Studies conducted by Pujiono *et al* [20] and Febrian and Figueredo [8] using RAG chatbots for the context of regulatory documents in Indonesia, particularly related to finance, show the potential for chatbots to become a decision-making tool. However, based on the literature review conducted, no specific research has been found that develops a Retrieval-Augmented Generation (RAG)-based chatbot to support the retrieval of SPBE supervision and evaluation guidelines that can assist assessors in conducting audits. In fact, the characteristics of SPBE evaluation, which are highly dependent on regulations, documentary evidence, and interpretation of indicators, indicate a high need for an information retrieval system that is accurate, verified, and contextual. This research gap is supported by national efforts to strengthen the implementation of SPBE through digital innovations that improve the efficiency and accuracy of evaluation [9].

Our research proposes the development of a RAG chatbot specifically for assessors to assist in the SPBE audit process based on the PANRB Ministerial Guideline No. 3 of 2024 concerning "Pedoman Tata Cara Pemantauan dan Evaluasi SPBE". The chatbot is expected to be able to provide relevant and accurate information that can be verified quickly, thereby improving the consistency of assessments, the efficiency of assessors' work, and the quality of improvement recommendations produced within the SPBE evaluation framework. Thus, this research provides practical contributions in supporting the consistency and efficiency of assessors' work, as well as methodological contributions in expanding the use of RAG in the context of E-Government audit and evaluation in Indonesia.CONCLUSION

This study demonstrates that SPBEBOT can effectively support SPBE assessors by providing regulation-grounded answers and structured maturity level assessments through a Retrieval-Augmented Generation approach. The evaluation results show strong performance in factual retrieval and contextual relevance, while audit-oriented questions can be addressed with correct maturity levels and domains. Quantitative evaluation using the RAGAS framework confirms high retrieval completeness and answer relevancy, although indicator, aspect, and level precision remains a challenge in scenarios involving overlapping assessment criteria.

The main contribution of this work is the application of a Vector RAG architecture for SPBE auditing, which improves information retrieval efficiency while preserving regulatory traceability. Nevertheless, the current vector-based approach has limitations in fully modeling the hierarchical nature of SPBE indicators, aspects, and levels, and the evaluation relied primarily on external assessors. Future study should explore more semantically structured retrieval methods and involve internal assessors to enhance validation depth and improve the reliability of audit-oriented decision support in SPBE evaluations.

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REFERENCE

[1] A. Mappasere, Fatmawati A., and A. Abdi, "The Role of E-government in Improving Efficiency and Transparency in Public Services in Indonesia," *KnE Social Sciences*, 2025.
[2] D. Fardiah, F. Darmawan, R. Rinawati, V. E. M. Supaat, and E. Hadnansyah, "Digital transformation through electronic-based government system performance as public relations strategy," *PRofesi Humas*, vol. 9, no. 1, pp. 23–48, Aug. 2024, doi: 10.24198/prh.v9i1.56095.
[3] Peraturan Presiden, "Peraturan Presiden Nomor 95 Tahun 2018," 2018.
[4] Menteri PANRB, "Indonesia Naik 13 Peringkat pada UN E-Government Survey 2024, Menteri PANRB: Kolaborasi Akselerasi Transformasi Digital Pemerintah melalui SPBE."
[5] Menteri PANRB, "Peraturan PANRB NO 59 Tahun 2020 Pemantauan dan Evaluasi SPBE," 2020.
[6] Menteri PANRB, "Pedoman Menteri PANRB Nomor 3 Tahun 2024 Pedoman Tata Cara Pemantauan dan Evaluasi SPBE," 2024.
[7] D. S. Putra, Selo, and S. Fauziati, "EVALUASI DOMAIN MANAJEMEN SPBE PEMERINTAH KOTA YOGYAKARTA BERDASARKAN PERATURAN MENTERI PAN-RB NOMOR 59 TAHUN 2020," *JIKO (Jurnal Informatika dan Komputer)*, vol. 5, pp. 54–61, 2022.
[8] G. F. Febrian and G. Figueredo, "KemenkeuGPT: Leveraging a Large Language Model on Indonesia's Government Financial Data and Regulations to Enhance Decision Making," Jul. 2024, [Online]. Available: http://arxiv.org/abs/2407.21459
[9] V. Rahma Aulia *et al.*, "Mapping SPBE Challenges in Indonesia: Systematic Review by Architecture Domain," *Jurnal Komputer Teknologi Informasi Sistem Informasi (JUKTISI)*, vol. 4, no. 1, pp. 174–183, Jun. 2025, doi: 10.62712/juktisi.v4i1.368.
[10] A. S. Asad and F. M. Khan, "Docu Mind AI – Intelligent Document Analysis System," *Indian Journal of Computer Science and Technology*, pp. 06–11, Sep. 2025, doi: 10.59256/indjcst.20250403002.
[11] M. Wondra, "Auditing Large Language Models," 2025.
[12] S. Emett, M. Eulerich, E. Lipinski, N. Prien, and D. A. Wood, "Leveraging ChatGPT for Enhancing the Internal Audit Process-A Real-World Example from a Large Multinational Company," 2023.
[13] J. Huang, H. Zhu, C. Xu, T. Zhan, Q. Xie, and J. Huang, "AuditWen:An Open-Source Large Language Model for Audit," Oct. 2024, [Online]. Available: http://arxiv.org/abs/2410.10873
[14] J. Mökander, J. Schuett, H. R. Kirk, and L. Floridi, "Auditing large language models: a three-layered approach," *AI and Ethics*, vol. 4, no. 4, pp. 1085–1115, Nov. 2024, doi: 10.1007/s43681-023-00289-2.
[15] L. Huang *et al.*, "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," *ACM Trans. Inf. Syst.*, vol. 43, no. 2, Jan. 2025, doi: 10.1145/3703155.
[16] Y. Zhang *et al.*, "Knowledge Overshadowing Causes Amalgamated Hallucination in Large Language Models," Jul. 2024, [Online]. Available: http://arxiv.org/abs/2407.08039
[17] Y. Gao *et al.*, "Retrieval-Augmented Generation for Large Language Models: A Survey," Dec. 2023, [Online]. Available: http://arxiv.org/abs/2312.10997
[18] L. Bahr, C. Wehner, J. Wewerka, J. Bittencourt, U. Schmid, and R. Daub, "Knowledge Graph Enhanced Retrieval-Augmented Generation for Failure Mode and Effects Analysis," Jun. 2024, [Online]. Available: http://arxiv.org/abs/2406.18114
[19] H. Tohir, N. Merlina, and M. Haris, "UTILIZING RETRIEVAL-AUGMENTED GENERATION IN LARGE LANGUAGE MODELS TO ENHANCE INDONESIAN LANGUAGE NLP," *JITK (Jurnal Ilmu Pengetahuan dan Teknologi Komputer)*, vol. 10, no. 2, pp. 352–360, Nov. 2024, doi: 10.33480/jitk.v10i2.5916.
[20] I. Pujiono, I. M. Agtyaputra, and Y. Ruldeviyani, "IMPLEMENTING RETRIEVAL-AUGMENTED GENERATION AND VECTOR DATABASES FOR CHATBOTS IN PUBLIC SERVICES AGENCIES CONTEXT," *JITK (Jurnal Ilmu Pengetahuan dan Teknologi Komputer)*, vol. 10, no. 1, pp. 216–223, Aug. 2024, doi: 10.33480/jitk.v10i1.5572.
[21] N. A. M. Herwanza, N. S. Harahap, F. Yanto, and F. Insani, "Penerapan Langchain Retriever dengan Model Chat Openai dalam Pengembangan Sistem Chatbot Hadis Berbasis Telegram," *JTIM : Jurnal Teknologi Informasi dan Multimedia*, vol. 6, no. 1, pp. 70–83, May 2024, doi: 10.35746/jtim.v6i1.514.
[22] A. Grattafiori *et al.*, "The Llama 3 Herd of Models," Nov. 2024, [Online]. Available: http://arxiv.org/abs/2407.21783
[23] Jerry Liu, "LlamaParse," 2024.
[24] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," May 2020, [Online]. Available: http://arxiv.org/abs/2005.11401
[25] S. Es, J. James, L. Espinosa-Anke, S. Schockaert, and E. Gradients, "RAGAS: Automated Evaluation of Retrieval Augmented Generation," 2024. doi: 10.18653/v1/2024.eacl-demo.16.
[26] B. Laugwitz, T. Held, and M. Schrepp, "Construction and evaluation of a user experience questionnaire," Springer Verlag, 2008, pp. 63–76. doi: 10.1007/978-3-540-89350-9_6.
[27] J. Jin *et al.*, "Hierarchical Document Refinement for Long-context Retrieval-augmented Generation."
[28] L. Zhang, Y. Wu, Q. Yang, and J.-Y. Nie, "Exploring the Best Practices of Query Expansion with Large Language Models," Jun. 2024, [Online]. Available: http://arxiv.org/abs/2401.06311
[29] R. Vega Centeno Olivera, A. M. de Souza, and J. C. dos Reis, "RFG Framework: Retrieval-Feedback-Grounded Multi-Query Expansion," in *International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management, IC3K - Proceedings*, Science and Technology Publications, Lda, 2025, pp. 509–516. doi: 10.5220/0013836900004000.
[30] J. Zhang, X. Liu, Y. Hu, C. Niu, F. Wu, and G. Chen, "RAGRouter: Learning to Route Queries to Multiple Retrieval-Augmented Language Models." [Online]. Available: https://github.com/OwwO99/RAGRouter.